

DCOS

Digital Construction Operating System

GFA, Site Area & Cost per m²

Measurement & Reporting Guideline for Quantity Surveyors

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Applies To	All QS Engineers, Estimators, and anyone producing a cost/m ² figure in DCOS
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1. Purpose & Rule of Use

Cost per square metre (\$/m²) is the most-used benchmark in construction: it sanity-checks a tender, compares projects, and is the number every client and director asks for first. But \$/m² is only meaningful if everyone measures the same way. Two QS engineers using different area conventions will report different \$/m² for the same building — and both will look wrong.

This guideline fixes ONE house convention for measuring Gross Floor Area (GFA) and Site Area, and ONE standard method for allocating cost to each, so every DCOS cost summary is comparable across projects, years, and staff.

The Governing Principle

GFA measures FLOOR PLATES, not cost items. Piles have cost but no floor — they divide by the building's GFA. A roof has area but no floor — its cost also divides by GFA. External works have neither building cost nor building floor — they divide by SITE AREA and never enter the building \$/m².

Rule of use: no cost/m² figure may be issued in a tender review, management report, or client document unless it is calculated per this guideline. Deviations require QS Manager approval recorded in writing.

2. Definitions

Term	Definition
GFA (Gross Floor Area)	Total area of all counted floor plates, measured to the OUTSIDE face of external walls, per the convention in Section 3. Includes internal walls, columns, circulation, and shafts.
GFA above ground	GFA of all floors at or above ground level, including counted roof-level rooms.
GFA basement	GFA of all floors below ground level. Always recorded separately from above-ground GFA.
GFA total	GFA above ground + GFA basement.
Site Area	Total land area of the project site within the legal boundary, per title/survey plan.
Building footprint	Ground-level area covered by the building. Used for site coverage checks — NOT a cost denominator.
Building cost	Contract cost of the building works only — excludes external works and land.
External works cost	Cost of works outside the building line: roads, boundary, drainage, landscape, guard house, site utilities.
Elemental cost/m ²	Cost of one trade/element (substructure, frame, finishes, MEP...) ÷ GFA total.
Blended \$/m ²	Total building cost ÷ GFA total (basement mixed in).
Split \$/m ²	Above-ground cost ÷ above-ground GFA reported separately from basement cost ÷ basement GFA.

3. GFA Measurement Convention (House Rules)

Measure each floor plate from the architectural floor plans to the OUTSIDE face of external walls. Then apply the inclusion table:

Element	Count in GFA	Rule
Typical enclosed floor	100%	Full plate to outside face of external walls
Basement floor	100%	Full plate — recorded as GFA basement, never mixed silently into above-ground
Covered balcony / open-sided corridor / covered walkway / canopy over entrance	50%	Half area — covered but not enclosed
Uncovered terrace, open roof deck, open parking on grade	0%	No enclosure — excluded
Lift shafts, stair cores, service risers	100%	Counted at EVERY floor they pass through
Double-height void (lobby, atrium, auditorium airspace)	Once	Counted once at the lowest floor; the void opening above counts 0%
Enclosed roof-level rooms (lift motor room, tank room, plant room, stair bulkhead)	100%	They are floor plates — count them in above-ground GFA
Roof structure, waterproofing, parapets	0%	Not a floor — cost divides by GFA total
Piles, pile caps, footings, ground beams	0%	Not a floor — cost shows as the Substructure elemental line ÷ GFA total
External works, boundary walls, site roads	0%	Different denominator — Site Area (Section 4)

Three Non-Negotiable Rules

1. GFA is ENTERED from drawings, never derived. Do not sum it from BOQ items or room areas — room-by-room totals miss walls, circulation, and shafts.
2. GFA is entered per LEVEL in DCOS and rolls up to building and project (Section 11). Parents are read-only.
3. When the design changes (added floor, changed footprint), the GFA is revised with a reason and the change is audit-logged. Historical \$/m² snapshots keep the GFA that was current at their time.

4. Site Area Convention

- Site Area = legal land area from the title deed / survey plan, entered once at the PROJECT node.
- If the contract covers only part of a larger site, use the defined works-area boundary from the contract drawings and note the source.
- Site Area is the denominator for external works \$/m² ONLY. It never divides building cost.
- Building footprint may be recorded for site-coverage ratio (footprint ÷ site area) but is not a cost denominator.

5. Cost Allocation Rules — What Divides by What

Cost	Denominator	Shown as
Building works total (incl. piling, roof, prelims share)	GFA total	Building \$/m ² (blended)
Above-ground building works	GFA above ground	Split reporting line 1
Basement works (excavation, retaining, waterproofing, basement structure)	GFA basement	Split reporting line 2
Substructure (piling, pile caps, ground beams)	GFA total	Elemental line inside building \$/m ²
Each trade/element (frame, finishes, MEP, prelims)	GFA total	Elemental \$/m ² table
External works	SITE AREA	External \$/m ² site — SEPARATE line, never inside building \$/m ²

Building \$/m² = Building cost ÷ GFA_total
 Elemental \$/m² = Element cost ÷ GFA_total
 Basement \$/m² = Basement cost ÷ GFA_basement
 External \$/m² (site) = External cost ÷ Site_Area
 NEVER: (Building + External) ÷ GFA as the headline building benchmark

A deep-piled project shows a HIGH substructure elemental line — that is the correct, visible signal. Hiding foundation cost by inflating area is the classic mistake this guideline exists to stop.

6. Case A — Building WITHOUT Basement

Project: 10-storey school building. Contract for building works only = \$3,595,200 (external works excluded, handled in Case C).

Step 1 — Measure GFA per level

Level	Plate area	Rule applied	Counted
Ground floor	850 m ²	100%	850
Levels 2–10 (9 floors)	9 × 850 m ²	100%	7,650
Roof: lift motor room + tank room	60 m ²	Enclosed roof rooms 100%	60
Roof: open deck	400 m ²	Uncovered 0%	0
GFA above ground = GFA total			8,560 m²

Step 2 — Compute the benchmark

Building \$/m² = 3,595,200 ÷ 8,560 = \$420.00 /m²

Step 3 — Elemental breakdown (÷ GFA total 8,560)

Element	Cost (USD)	\$/m ²
Substructure (piling, pile caps, ground beams)	428,000	50.00
Superstructure (concrete frame)	1,112,800	130.00
Architectural works & finishes	1,027,200	120.00
MEP services	856,000	100.00
Preliminaries	171,200	20.00
Building total	3,595,200	420.00

Note how the piling cost is fully visible as the \$50/m² substructure line — divided by the whole GFA, contributing zero area of its own.

7. Case B — Building WITH Basement

Same school, now with one basement level of 850 m² (parking + plant). Basement works (excavation, retaining, waterproofing, basement structure) = \$722,500. Illustrative — in reality substructure design also changes; this example isolates the area logic.

Step 1 — GFA split

Component	Area
GFA above ground (Case A)	8,560 m ²
GFA basement (1 × 850)	850 m ²
GFA total	9,410 m²

Step 2 — Report BOTH ways (mandatory)

Reporting line	Calculation	Result
Above-ground \$/m ²	$3,595,200 \div 8,560$	420.00
Basement \$/m ²	$722,500 \div 850$	850.00
Blended building \$/m²	$(3,595,200 + 722,500) \div 9,410 = 4,317,700 \div 9,410$	458.84

Why split reporting is mandatory

The basement costs 850 \$/m² — roughly double a typical floor. If you publish only the blended 458.84, this project looks 9% more expensive than the no-basement school at 420.00, and someone will wrongly conclude the building got costlier. Split reporting shows the truth: the above-ground building is IDENTICAL at 420.00; the basement is a separate, expensive component. Blended is for portfolio comparison; split is for understanding.

8. Case C — External Works

Site area 4,500 m² (title deed). External works: boundary wall, internal road & parking, storm drainage, landscaping, guard house = \$180,000.

External works \$/m² (site) = $180,000 \div 4,500 = \$40.00$ /m² of site
 WRONG: adding 180,000 into the building numerator
 $(4,317,700 + 180,000) \div 9,410 = 477.97$ ← NOT the building benchmark

- External works relate to the land, not the floor plates. A small building on a large site would be unfairly penalized if external cost divided by GFA.
- The 477.97 all-in figure MAY be shown as a memo line labelled “whole project ÷ GFA” — but never as “building \$/m²”, and never used to benchmark against building-only figures.

9. Case D — Combined Project & the Standard Final Cost Summary

The full project = Case B building + Case C external works. Total contract \$4,497,700. Every project cost summary issued from DCOS must follow this exact format:

STANDARD FINAL COST SUMMARY (mandatory format)

No.	Description	Cost (USD)	Denominator	\$/m ²
1	Building works — above ground	3,595,200	8,560 m ² GFA	420.00
2	Building works — basement	722,500	850 m ² GFA	850.00
A	Building subtotal (blended)	4,317,700	9,410 m² GFA	458.84
3	External works	180,000	4,500 m ² site	40.00
B	TOTAL CONTRACT	4,497,700	—	—
memo	Whole project ÷ GFA (memo only — not a benchmark)	4,497,700	9,410 m ² GFA	477.97

Below the summary, attach the elemental table (Section 6 Step 3 format) computed on GFA total, and state the GFA convention reference: “Areas per DCOS-QS-GDL-001 V1.0”. Client-facing versions show sell-side costs only, per the firewall rule (SOP-QS-05A / RPT-001 G7).

Reading the summary — what each line answers

Line	Question it answers
Line 1 (420.00)	Is this building's construction economical? Compare with other above-ground buildings of the same type.
Line 2 (850.00)	What did going underground cost us? Compare basement decisions across projects.
Line A (458.84)	Portfolio blended rate — for company-wide \$/m ² statistics.
Line 3 (40.00)	Site development intensity — compare against site area, not building size.
Memo (477.97)	Answering a director who asks “total divided by area” — labelled so it is never mistaken for the building benchmark.

10. Common Mistakes & Validation Checklist

The five classic mistakes

1. **Counting foundations or roof structure as area.** They are cost, not floor plate — 0% area, full cost ÷ GFA total.
2. **Blending the basement silently.** Always report split + blended; a blended-only figure misleads every comparison.
3. **Dividing external works by GFA.** External works ÷ Site Area, always.
4. **Deriving GFA from BOQ or room areas.** GFA is measured from floor plans to the outside face — room sums miss walls, shafts, circulation.
5. **Changing GFA without revision control.** A GFA edit silently changes every historical-looking benchmark. Revise with reason; snapshots keep their original GFA.

Checklist before issuing any \$/m² figure

- GFA entered per level, from the current drawing revision, convention table applied and noted.
- Basement levels flagged; above/below split available.
- Site area entered at project node with source (title deed / survey ref).
- External works WBS branch excluded from building numerator.
- No level has cost > 0 with GFA = 0 (DCOS warns — do not ignore the warning).
- Summary follows the Section 9 mandatory format; convention reference cited.
- Client version contains sell-side figures only (firewall check).

11. Entering the Data in DCOS

1. Open the WBS tree → select each **Level** node → Quantities panel → add metric GFA with value (m²), source = drawing revision reference.
2. Basement levels: ensure the node is typed/flagged as below-ground so the split reports automatically.
3. Roof-level enclosed rooms: enter their 60 m² (etc.) as GFA on the roof level node — the open deck is simply not entered.
4. Select the **Project** node → add metric SITE_AREA with the title-deed area and source reference.
5. Building and project GFA totals roll up automatically — parent values are read-only.
6. Design change: edit the level GFA → the system requires a reason and logs the change; existing S-curve/EVM snapshots retain their original GFA.
7. The QS Dashboard “Cost/m²” card and the Final Cost Summary report (RPT series) read ONLY these entered values — if the card shows “no area data”, the input above is missing.

System reference: metrics are stored in wbs_node_quantities (metric_code GFA / SITE_AREA); feature specification FS-QS-012 in DDS-001 V2.0.

12. Approval & Revision

The 50% balcony treatment, void rules, and split-reporting mandate are HOUSE DECISIONS. They bind all staff once approved below. Amendments follow document control (new version, reason, re-approval).

Role	Name	Signature	Date
QS Manager (convention authority)			
Commercial Director (approver)			

Version	Date	Description	Author
V1.0	2026-07-16	Initial issue — GFA/site conventions, cost allocation rules, worked Cases A–D, standard final cost summary format, DCOS entry procedure	DCOS Commercial Team

End of Document — DCOS-QS-GDL-001 V1.0